

DATA ANALYSIS INTERNSHIP

DAY 9 REPORT

Student Performance Analysis using Pandas and Matplotlib

Introduction

On Day 7, a **Student Performance dataset from Kaggle** was analyzed using **Pandas in Google Colab**. The dataset included information related to **student study time, subject marks, grades, gender, and email details**, consisting of **1499 records and 12 columns**. Various dataframe operations, statistical analysis, filtering methods, and visualization concepts were explored. In addition, **Matplotlib** was introduced for creating graphs and visual representations of data.

1. Dataset Import

The student dataset was imported into a Pandas dataframe for analysis.

Function Used:

```
pd.read_csv()
```

Purpose:

Used for reading CSV datasets into a dataframe environment.

2. Dataset Overview

Several dataframe functions were used to inspect and understand the dataset.

Functions Practiced:

- `head()`
- `tail()`
- `shape`
- `columns`
- `dtypes`

- `info()`
- `describe()`

Purpose:

These functions helped in:

- Viewing dataset rows and columns
 - Identifying column data types
 - Understanding dataframe structure
 - Generating statistical summaries
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3. Missing Value Analysis

The dataframe was checked for missing or empty values.

Function Used:

`df.isnull()`

Purpose:

Identifies null or missing values in the dataset.

4. Statistical Analysis

Several statistical functions were applied to student subject marks.

Functions Used:

- `sum()`
- `mean()`
- `mode()`
- `max()`
- `min()`
- `count()`
- `std()`

Tasks Performed:

- Calculated total mathematics score
- Found average self-study hours per week
- Identified the most common gender

- Determined maximum history marks
 - Found minimum physics marks
 - Counted total student IDs
 - Measured standard deviation of geography scores
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5. Sorting and Duplicate Management

Sorting and duplicate removal techniques were practiced.

Functions Used:

- `sort_index()`
- `drop_duplicates()`

Purpose:

Used for arranging dataframe indexes and removing repeated records.

6. Dataframe Column Operations

Different column modification techniques were explored.

Functions Used:

- `drop()`
- `rename()`
- `insert()`
- `replace()`
- `astype()`

Activities Performed:

- Removed unnecessary columns such as Email
 - Modified column names
 - Added a new Age column
 - Updated dataframe values
 - Converted datatypes into float format
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7. Filtering and Query Methods

Filtering was performed to retrieve specific student information.

Functions Used:

- `query()`
- Conditional dataframe indexing

Examples:

- Students scoring above 90 in Mathematics
- Students scoring above 80 in History
- Students with English marks equal to 90

Purpose:

Helps in extracting records based on conditions.

8. Random Sampling

Random records were selected from the dataframe.

Function Used:

`df.sample(1)`

Purpose:

Displays randomly selected rows from the dataset.

9. Indexing Techniques

Index-based operations were applied to the dataframe.

Functions Used:

- `loc[]`
- `set_index()`

Purpose:

Used for:

- Accessing rows through labels
 - Setting the Id column as the dataframe index
-

10. Grouping Data

Grouping operations were performed for aggregation.

Function Used:

```
df.groupby("Id").sum()
```

Purpose:

Groups related records and applies calculations collectively.

11. Dataset Combination using Merge

Datasets were combined using a common column.

Function Used:

```
pd.merge(df, df2, on="Id")
```

Purpose:

Merges two datasets based on matching column values.

12. Conversion to NumPy Array

The dataframe was converted into NumPy format.

Function Used:

```
df.to_numpy()
```

Purpose:

Converts dataframe values into array format for numerical operations.

13. Text Processing Functions

String operations were applied to textual data.

Functions Practiced:

- `upper()`
- `capitalize()`
- `len()`

- `contains()`
- `startswith()`

Purpose:

Useful for text formatting and data cleaning.

14. Pandas Series Creation

A Series object was created.

Function Used:

```
pd.Series([df])
```

Purpose:

Used to generate a Pandas Series for storing one-dimensional data.

15. Counting Unique Values

Unique values in columns were identified.

Function Used:

```
df.nunique()
```

Purpose:

Counts distinct values present in each column.

Introduction to Matplotlib

Definition

Matplotlib is a **Python visualization library** used to create charts, graphs, and plots for better understanding of datasets.

Import Statement

```
import matplotlib.pyplot as plt
```

Purpose of Matplotlib

- Creating visual charts and plots
 - Representing data graphically
 - Comparing different values
 - Identifying patterns and trends
 - Supporting data analysis and machine learning
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Common Matplotlib Functions Practiced

Function	Purpose
<code>plot()</code>	Creates line graph
<code>bar()</code>	Creates vertical bar chart
<code>barh()</code>	Creates horizontal bar chart
<code>scatter()</code>	Displays scatter plot
<code>hist()</code>	Creates histogram
<code>pie()</code>	Creates pie chart
<code>xlabel()</code>	Adds x-axis label
<code>ylabel()</code>	Adds y-axis label
<code>title()</code>	Adds graph title
<code>legend()</code>	Displays labels
<code>grid()</code>	Adds grid lines
<code>show()</code>	Displays graph

<code>figure()</code>	Creates figure window
<code>subplot()</code>	Creates multiple plots
<code>savefig()</code>	Saves graph as image

Conclusion

On Day 7, I gained practical experience in **Pandas and Matplotlib** for data analysis and visualization. Different dataframe operations such as **dataset inspection, filtering, grouping, merging, indexing, and statistical analysis** were performed using a student performance dataset. I also explored **Matplotlib functions for generating graphs and charts**, which helped in understanding data patterns visually. This session enhanced my knowledge of **data manipulation, analysis, and visualization techniques in Python**.